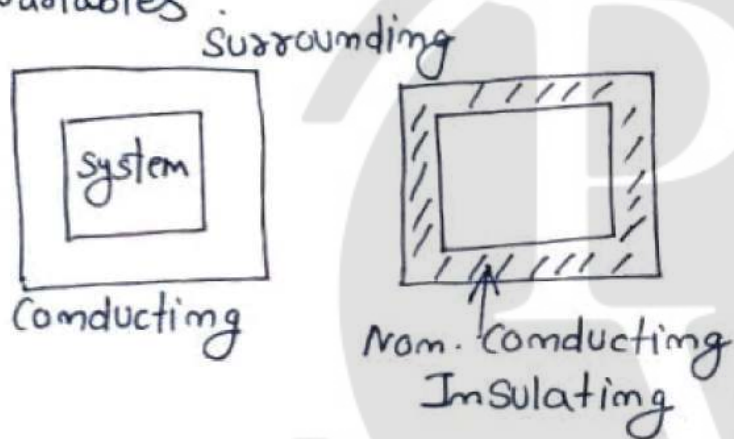


THERMODYNAMICS

⇒ Chapter Overview =

Thermodynamics is the branch of physics that deals with the concepts of Heat, work, and energy, and their interconversion. It studies macroscopic systems and establishes relationships between thermodynamic variables.



⇒ Key Concepts =

① Charles's law = when pressure is constant then the volume of the gas of a definite quantity is directly proportional to its temperature.

$$V \propto T \quad \left[\frac{V}{T} = \text{Constant} \right]$$

② Boyle's law = When temperature is constant then the pressure of given mass of a perfect gas is inversely proportional to its volume.

$$P \propto \frac{1}{V}$$

$$PV = \text{Constant}$$

Ideal gas equation = $\frac{PV}{T} = \text{Constant}$

$$\frac{PV}{T} = R$$

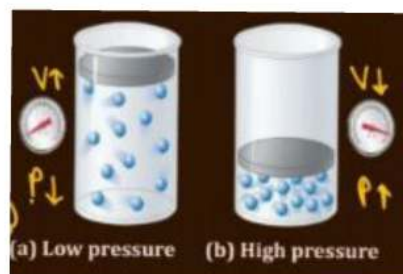
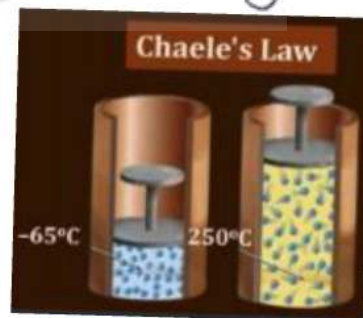
R is a Universal gas constant $R = 8.31 \text{ Joule/molK}$

⇒ Thermodynamic terms =

→ System = The part of the Universe under observation (eg. gas in a cylinder)

→ Surroundings = Everything outside the system.

→ Boundary = The surface separating the system and surroundings.



① Open System = Exchanges energy and matter (e.g. boiling water)

② Closed System = Exchanges energy but not matter (e.g. gas in a sealed cylinder)

③ Isolated System = Exchange neither energy nor matter (e.g. an insulated thermos)

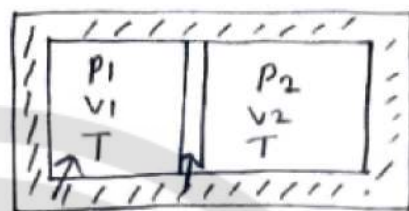
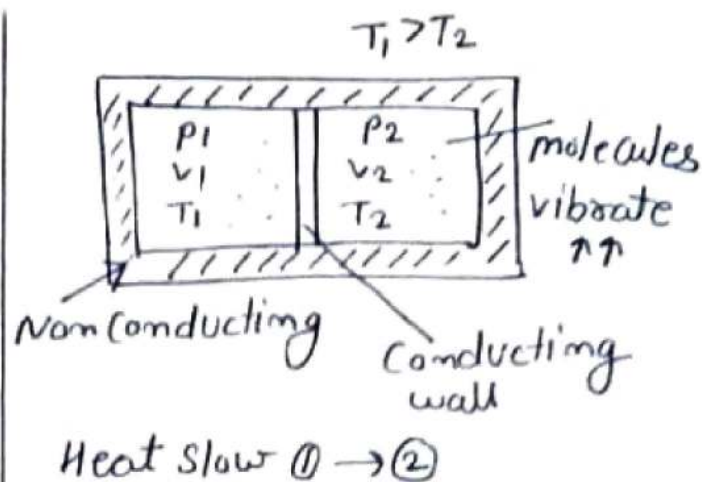


⇒ Thermodynamics Equilibrium

The system whose all macroscopic properties (P, V, T) are constant with time then we can say that the system is in thermal equilibrium.

→ That means there is no exchange of macroscopic variables with surrounding.

→ Can a system be in thermal equilibrium with surrounding at all temperature.

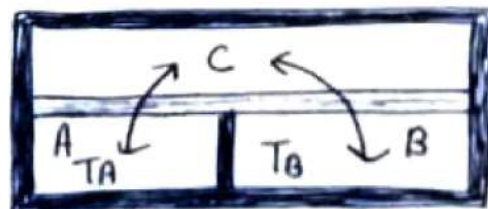


Thermal Equilibrium

⇒ Zeroth law of thermodynamic

If two systems are in thermal equilibrium with a third system, they are in thermal equilibrium with each other.

→ Implication: Defines Temperature as a measurable quantity.



$$T_A \neq T_B$$

$$\Rightarrow T_A = T_C$$

$$\Rightarrow T_B = T_C$$

That $T_A = T_B$

$$T_A = T_B = T_C$$

⇒ Thermodynamic State Variables = state variable

are quantities that define the state of a system

examples include:

→ pressure (p)

→ volume (v)

→ Temperature (T)

→ Internal Energy (U)

→ These are macroscopic properties that describe a system's equilibrium.

⇒ Mechanical Equivalent of work and Heat = According

to Joule's work done on a system in proper condition w and Heat obtained H , the are proportional to each other.

→ J is called mechanical Equivalent of Heat.

1 Calorie = 4.2 Joule

1 Joule = 0.24 cal

w & H

$$w = JH$$

Joule's law Heating

where ' J ' is called mechanical Equivalent of Heat.

⇒ Work, Heat and Internal energy =

Heat = The flow of energy due to temperature difference is called heat while the flow of energy without temperature difference is called mechanical work.

→ Heat and work are not position variable in thermodynamics.

→ Internal energy depends upon the final and Initial state of the system, not on the path.

→ SI unit = Joule (J)

practical = Calorie (cal)

flow = [High Temp → Low Temp]

→ for Increase in Temperature

$$\Delta Q = m S \Delta T$$

$$\Delta Q = n C \Delta T$$

n = no. of moles

C = molar Specific Heat

(i) at constant pressure

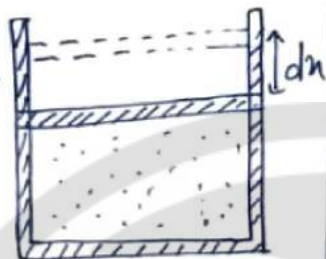
$$\Delta Q_p = n C_p \Delta T$$

(ii) At constant volume

$$\Delta Q_v = n C_v \Delta T$$

→ Work = Let a gas in a container, whose piston is frictionless and has negligible mass, on giving heat to the gas from base of a container, gas expands by 'dn'

→ work done to displace piston by 'dn'



$$dw = Fdn \quad \left[\begin{array}{l} \text{pressure} = F/A \\ F = AP \end{array} \right]$$

$$dw = PAdn \quad \text{--- (1)}$$

Change in volume is $dv \Rightarrow dv = Adn$

$$dw = Pdv$$

Integrate both sides

$$\int dw = \int Pdv$$

$$w = \int Pdv$$

→ Area of pressure and volume graph, will represent work.

→ If volume change from v_1 to v_2

$$w = \int_{v_1}^{v_2} Pdv$$

$$w = P \int_{v_1}^{v_2} dv$$

$$w = P [v]_{v_1}^{v_2}$$

$$w = P [v_2 - v_1]$$

$$w = P\Delta v$$

→ Initial volume v_1

Final volume v_2

$$\Delta v = v_2 - v_1$$

(i) If $\Delta v \oplus ve$ when $v_2 > v_1$

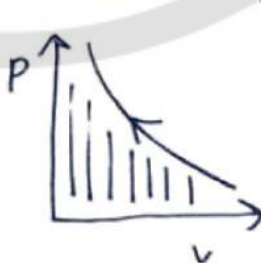
Then $w = \oplus ve$ Expansion
work done by system = $\oplus ve$

(ii) work done by surrounding or
work done on system = $\ominus ve$ Compression

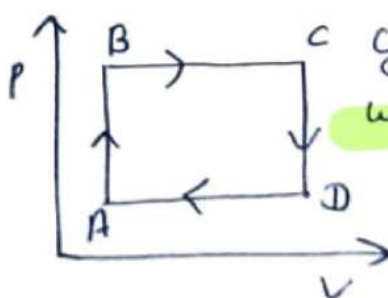
→ work done depends on path



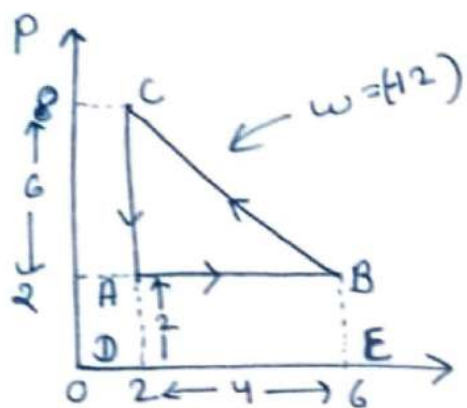
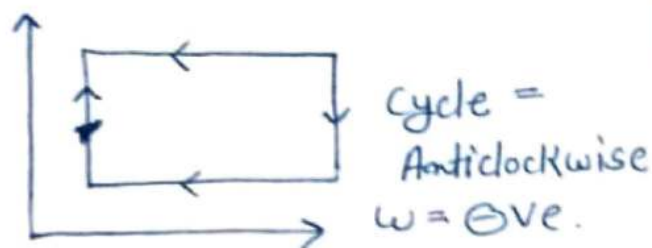
$v \uparrow \uparrow \rightarrow \downarrow p$
 $w = \oplus ve$



$v \downarrow \downarrow \rightarrow \uparrow p$
 $w = \ominus ve$



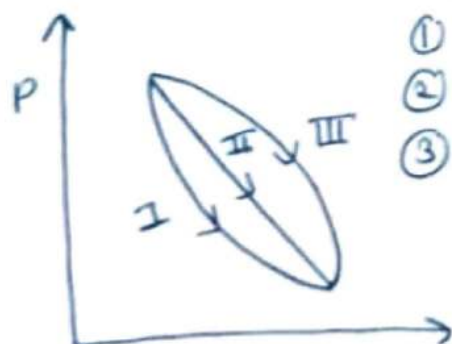
Cycle = clockwise
 $w = \oplus ve$



(i) $W(A \rightarrow B)$
 = Area of $ABED$
 = $4 \times 2 = 8 \text{ unit}$

(ii) $W(B \rightarrow C)$
 = Area of $\Delta + \text{Area of Rectangle}$
 = $(\frac{1}{2} \times 4 \times 6) + 8$
 = $12 + 8$
 = -20

(iii) $W(C \rightarrow A)$
 $W = 0$
 $\Delta V = 0$



- (1) $W_1 > W_2 > W_3$
 (2) $W_1 = W_2 = W_3$
 (3) $W_1 < W_2 < W_3$

Area $\uparrow \uparrow$ (3)
 $[W \uparrow \uparrow]$
 $[Q \uparrow \uparrow]$

→ Internal Energy =

The store energy of a system due to which without the external energy the system is capable of doing work is called Internal energy.

→ The energy possessed by due to molecular motion and molecular configuration

(a) Internal Kinetic energy

The Energy due to molecular motion is called internal kinetic energy.

→ It depends upon temperature

$$T \uparrow \rightarrow KE \uparrow \uparrow$$

(b) Internal potential energy

The Energy due to molecular configuration is called Internal kinetic energy.

→ It depends upon the distance between the molecules and attraction forces.

Internal energy U

$$U = K.E. + P.E.$$

⇒ If no change in volume

$$\Delta w = 0$$

$$\Delta Q = \Delta U \quad [\Delta Q = nC_v \Delta T]$$

$$\boxed{nC_v \Delta T = \Delta U}$$

→ pressure Constant [Isobaric]

$$\Delta Q = \Delta U + \Delta w$$

$$\boxed{\Delta Q = nC_p \Delta T}$$

⇒ FLT D =

$$\Delta Q = \Delta U + \Delta w$$

$$0 = nC_v \Delta T + P \Delta V$$

$$0 = nC_v dT + P dV$$

put the value of dT from equation — ①

$$0 = nC_v \left(\frac{P dV + V dP}{R} \right) + P dV$$

$$0 = \frac{nC_v (P dV + V dP) + P dV R}{R}$$

$$0 = n(C_v P dV + C_v V dP + P dV R)$$

$$0 = C_v P dV + C_v V dP + P dV R \quad [n=1]$$

$$0 = P dV [C_v + R] + V dP C_v$$

$$0 = P dV C_p + V dP C_v \quad \left[\begin{array}{l} \text{Mayer's relation} \\ C_p = C_v + R \end{array} \right]$$

→ divide by $C_v P V$

$$\Rightarrow \frac{P dV C_p}{C_v P V} + \frac{V dP C_v}{C_v P V} = 0$$

$$= \left(\frac{C_p}{C_v} \right) \frac{dV}{V} + \frac{dP}{P} = 0$$

we know that $\left(\frac{C_p}{C_v} = \gamma \right)$

$$\gamma \frac{dV}{V} + \frac{dP}{P} = 0$$

Integrating both side

$$\gamma \log V + \log P = \text{Constant}$$

$$\log V^\gamma + \log P = K$$

$$\log [P V^\gamma] = K$$

$$P V^\gamma = e^K$$

$$\boxed{P V^\gamma = K} \quad \text{Ideal gas equation for adiabatic process}$$

⇒ Work done in a adiabatic process =

$$w = \int P dV \quad \text{--- ①}$$

Let the volume change from V_1 to V_2

Ideal gas equation for adiabatic

$$P V^\gamma = K$$

$$P = \frac{K}{V^\gamma}$$

put the value of P in equation — ①

$$w = \int_{V_1}^{V_2} \frac{K}{V^\gamma} dV$$

$$w = K \int_{V_1}^{V_2} \frac{1}{V^\gamma} dV$$

$$w = K \int_{V_1}^{V_2} V^{-\gamma} dV$$

$$w = K \left[\frac{V^{-\gamma+1}}{-\gamma+1} \right]_{V_1}^{V_2}$$

$$w = \frac{K}{-\gamma+1} \left[V^{-\gamma+1} \right]_{V_1}^{V_2}$$

$$w = \frac{K}{-\gamma+1} \left[V_2^{-\gamma+1} - V_1^{-\gamma+1} \right]$$

Work done in Isothermal process

= Let the volume change of system from v_1 to v_2 , then work done.

$$\Rightarrow w = \int_{v_1}^{v_2} p dv \quad \text{--- (1)}$$

Ideal gas equation

$$pV = nRT$$

$$p = \frac{nRT}{V}$$

put the value in equation (1)

$$\Rightarrow w = \int_{v_1}^{v_2} \frac{nRT}{V} dv$$

$$\Rightarrow w = nRT \int_{v_1}^{v_2} \frac{1}{V} dv$$

$$\Rightarrow w = nRT [\log_e V]_{v_1}^{v_2}$$

$$w = nRT [\log_e v_2 - \log_e v_1]$$

$$w = nRT \log_e \left(\frac{v_2}{v_1} \right)$$

$$w = 2.303 nRT \log_{10} \left(\frac{v_2}{v_1} \right)$$

$$pV = \text{constant}$$

$$p \propto \frac{1}{V}$$

$$\frac{p_1}{p_2} = \frac{v_2}{v_1}$$

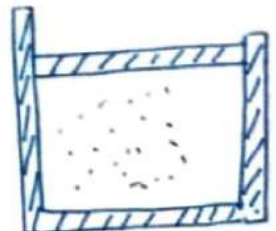
$$w = 2.303 nRT \log_{10} \left[\frac{v_2}{v_1} \right]$$

$$w = 2.303 nRT \log_{10} \left(\frac{p_1}{p_2} \right)$$

→ Adiabatic process = If there is no exchange of heat of the system with surrounding then system will be adiabatic process.

Condition =

(i) process should be very fast



(ii) walls should be perfectly insulating

First law of thermodynamic

$$\Delta Q = \Delta U + \Delta W$$

$$\Delta Q = 0 \text{ for adiabatic}$$

$$0 = \Delta U + \Delta W$$

$$\Delta W = -\Delta U$$

#2. Ideal gas equation

Ideal gas equation for n mole gas

$$pV = nRT$$

For 1 mole gas ($n=1$)

$$pV = RT$$

→ differentiating the above equation

$$p dv + v dp = R dT$$

$$\frac{p dv + v dp}{R} = dT \quad \text{--- (1)}$$

→ from FLT D

$$\Delta Q = \Delta U + \Delta W$$

$$W = \frac{1}{-\gamma+1} [K v_2^{-\gamma+1} - K v_1^{-\gamma+1}]$$

$$\left[p v^\gamma = K \right. \\ \left. p_1 v_1^\gamma = p_2 v_2^\gamma = K \right]$$

$$W = \frac{1}{-\gamma+1} [p_2 v_2^{\gamma-\gamma+1} - p_1 v_1^{\gamma-\gamma+1}]$$

$$W = \frac{1}{-\gamma+1} [p_2 v_2^{\gamma-\gamma+1} - p_1 v_1^{\gamma-\gamma+1}]$$

$$W = \frac{1}{-\gamma+1} [p_2 v_2 - p_1 v_1]$$

$$W = \frac{1}{\gamma-1} [p_1 v_1 - p_2 v_2]$$

$$W = \frac{1}{\gamma-1} [n R T_1 - n R T_2]$$

$$W = \frac{n R}{\gamma-1} (T_1 - T_2)$$

⇒ Slope of adiabatic = Ideal gas equation for adiabatic

$$p v^\gamma = K$$

Differentiation both side

$$v^\gamma dp + \gamma v^{\gamma-1} dv p = 0$$

$$v^\gamma dp = -\gamma p v^{\gamma-1} dv$$

$$\frac{dp}{dv} = \frac{-\gamma p v^{\gamma-1}}{v^\gamma}$$

$$\frac{dp}{dv} = -\gamma p v^{\gamma-1-\gamma}$$

$$\frac{dp}{dv} = -\gamma p v^{-1}$$

$$\frac{dp}{dv} = -\gamma p v^{-1}$$

$$\frac{dp}{dv} = -\gamma \frac{p}{v}$$

⇒ Slope of Isothermal and adiabatic process =

$$(m)_{\text{Isothermal}} \Rightarrow \frac{dp}{dv} = -\frac{p}{v}$$

Slope of Adiabatic process

$$(m)_{\text{adiabatic}} = \frac{dp}{dv} = -\gamma \frac{p}{v}$$

$$m_{\text{adiabatic}} = -\gamma \frac{p}{v}$$

$$m_{\text{adiabatic}} = \gamma m_{\text{Isothermal}}$$

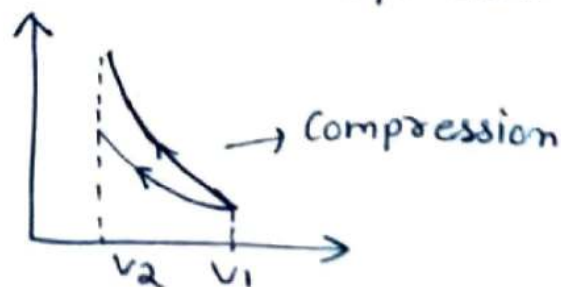
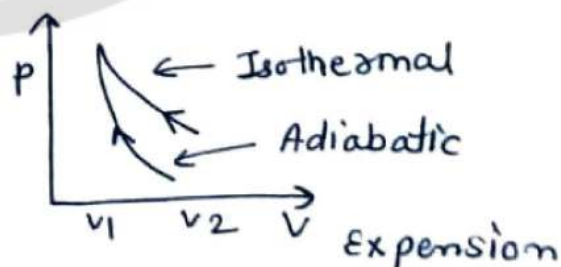
Slope of adiabatic is γ times the slope of Isothermal

$$\frac{m_{\text{adiabatic}}}{m_{\text{Isothermal}}} = \gamma$$



① → Adiabatic

② → Isothermal

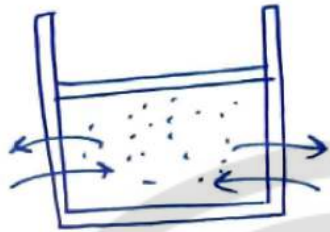


④ Isochoric process

- (a) Volume remains constant
- (b) work done.

→ Isothermal processes =

The process in which at constant temperature, p and v can change



→ Condition =
[during the experiment]

- ① The process should be slow
- ② wall should be perfectly conducting

→ Ideal gas equation

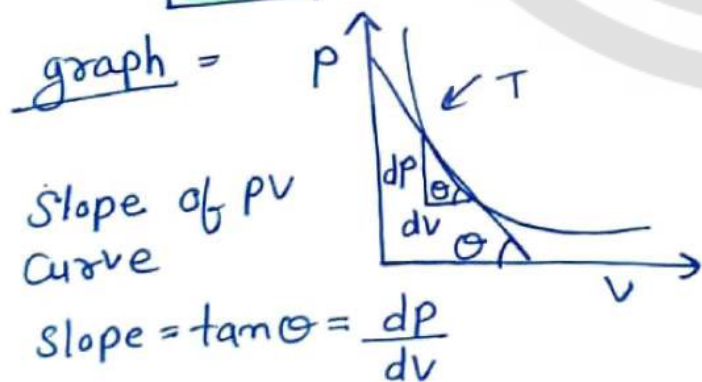
$$PV = nRT \quad \text{--- ①}$$

for Isothermal process

$$T = \text{Constant}$$

$$\Rightarrow PV = \text{Constant}$$

$$P \propto \frac{1}{V}$$



→ Slope of PV Curve =

Ideal gas equation for Isothermal

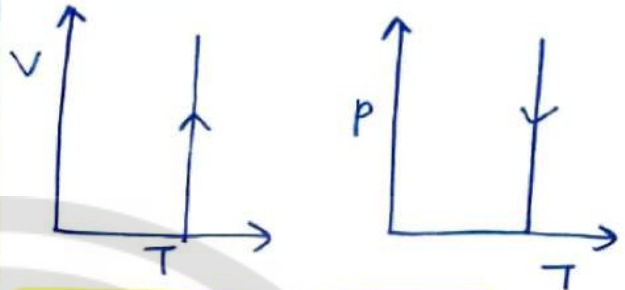
$$PV = \text{Constant}$$

① If both side

$$Pdv + vdp = 0$$

$$vdp = -Pdv$$

$$\boxed{\frac{dp}{dv} = -\frac{P}{V}}$$



→ Specific Heat for Isothermal =

$$\Delta Q = mc\Delta T$$

$$c = \frac{\Delta Q}{m\Delta T} \quad \Delta T = 0$$

$$\boxed{c = \infty}$$

→ Internal energy

$$T = \text{Constant}$$

$$\Delta T = 0$$

$$\text{Then } \Delta U = 0$$

no change in internal energy

FLTD =

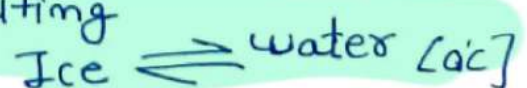
$$\Delta Q = \Delta U + \Delta W \quad [\Delta U = 0]$$

$$\Delta Q = \Delta W$$

$$\Delta Q = P\Delta V$$

Example = at constant Temperature

melting



→ Internal Energy does not depend on path,

It depends only on final and initial ~~path~~ point

→ In Ideal gas Total internal energy

$$U = K.E. + P.E.$$

$$U = KE + PE$$

$$U = KE$$

⇒ First Law of thermodynamics

"The value of heat given to a system is equal to the sum of change in internal energy and work done by the system."

$$\Delta Q = \Delta U + \Delta W$$

$$\Delta Q = \Delta U + P\Delta V$$

→ Energy cannot be created or destroyed but it can be converted from one form to another.

This law is based on the

$$\Delta Q = \Delta U + \Delta W$$

where =

ΔQ = Heat supplied to the system

ΔU = Change in internal energy

W = work done by the system

① The direction of the process is not decided

② This law does not give any information about the thermal sources i.e. whether the system is hot or cold.

③ The complete heat taken from the source by an engine cannot be completely converted into mechanical work if there is no change in internal energy. This question is also not answered by the first law.

⇒ Thermodynamic processes

① Isothermal process

(a) The Temperature remains constant

(b) work done.

② Adiabatic process

(a) No Heat Exchange

(b) work done

③ Isobaric process

(a) pressure remains constant

(b) work done.

→ Mayer's Relation =

first law of thermodynamic

$$\Delta Q = \Delta U + \Delta W \text{ --- (1)}$$

Let C_p and C_v are molar specific heat at constant pressure and volume respectively.

$$\Delta Q = n C_p \Delta T$$

$$C_p - C_v = R$$

$$\Delta U = n C_v \Delta T \quad \text{Mayer's Relation}$$

$$\Delta W = P \Delta V$$

put the values in equation (1)

$$n C_p \Delta T = n C_v \Delta T + P \Delta V$$

$$n C_p \Delta T = n C_v \Delta T + n R \Delta T$$

$$n \Delta T C_p = n \Delta T [C_v + R]$$

$$C_p = C_v + R$$

Isoobaric process =

$P = \text{constant}$

→ Ideal gas equation

$$PV = nRT \quad \left| \begin{array}{l} V \propto T \\ \Delta Q = n C_p \Delta T \end{array} \right.$$

$$V = \left(\frac{nR}{P} \right) T \quad \left| \begin{array}{l} V \propto T \\ \Delta Q = n C_p \Delta T \end{array} \right.$$

→ FLT D

$$\Delta Q = \Delta U + \Delta W$$

$$\Delta Q = \Delta U + \Delta W$$

Isochoric process =

$V = \text{constant}$

Ideal gas equation

$$PV = nRT$$

$$P = \left(\frac{nR}{V} \right) T$$

$$P \propto T$$

$$\Delta V = 0$$

$$W = 0$$

FLT D

$$\Delta Q = \Delta U + \Delta W$$

$$\Delta Q = \Delta U$$

$$\Delta Q = n C_v \Delta T$$

$$n C_v \Delta T = \Delta U$$

→ Cyclic process = The process in which system return to its Initial state through various changes.

Reversible and Irreversible process =

→ The process in which system return back with small changes in external condition.

→ There is no loss of energy

→ It is an ideal concept

→ Energy dissipating elements friction, resistance etc. should be absent.

→ does not Heat loss by Conduction Convection and radiation.

→ Temperature difference should be very small
Ex = Vapour Convert into water.

Irreversible process =

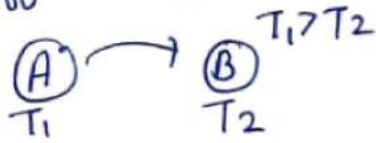
The process which cannot be return back.

→ All the natural processes in nature are irreversible

→ production of Heat due to friction or Heat produced in resistance with the

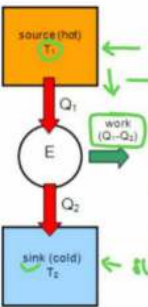
flow of current are also irreversible.

→ Diffusion of liquids and gases.



Heat engine =

$$\text{efficiency } (\eta) = \frac{\text{work done}}{\text{Heat given}}$$



$$\eta = \frac{Q_1 - Q_2}{Q_1}$$

$$\eta = 1 - \frac{Q_2}{Q_1}$$

$$\text{If } Q_2 = 0 \\ \eta = 1$$

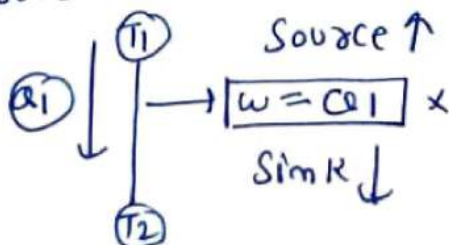
Second law of thermodynamics

→ It explains the direction of natural processes and introduces the concept of entropy.

→ Kelvin-Planck statement =

It is impossible to construct a heat engine that operates in a cycle and extracts heat from a single reservoir to do an equivalent amount of work.

→ A working substance cannot convert all the heat extracting from a source into mechanical work.



→ Clausius Statement =

Heat cannot spontaneously flow from a colder body to a hotter body, without any work being done on the system.

Thermodynamic process =

first law does not explain the process is possible or not.

→ All work can be converted into heat but all heat cannot be converted into work.

→ Kinetic energy can be converted into heat energy but heat energy cannot be converted into kinetic energy.

→ Efficiency of any engine cannot be 100%. $\eta \neq 100\%$.

